

Ismail Elyan

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience designing and building AI solutions specializing in Large Language Models (LLMs), LLMops, Retrieval-Augmented Generation (RAG), and agentic AI workflows. Proficient in building NLP and generative AI applications using LangChain, LangGraph, Hugging Face, PyTorch, and n8n for orchestration and automation. Strong problem-solving, analytical thinking, and research skills with a focus on scalable, production-ready AI systems. Transitioning toward MLOps and advanced LLMops practices with practical AWS deployment experience.

WORK EXPERIENCE

ML Trainee – Digital Egypt Pioneers Initiative (DEPI)

Oct 2024 – Jun 2025

- Completed a 9-month intensive AI and Machine Learning program covering NLP, Computer Vision, and end-to-end ML pipelines.
- Built, trained, and evaluated 10+ NLP and Computer Vision models using Python and PyTorch, achieving up to 91% accuracy on benchmark datasets.
- Designed full ML pipelines including data preprocessing, feature engineering, model evaluation, and deployment.
- Deployed models on Azure and Hugging Face Spaces, reducing inference latency by 25% through optimization techniques.

NLP Trainee – National Telecommunication Institute (NTI)

Aug 2025

- Completed 120 hours of intensive NLP training, achieving a 98% final score — top performer in the cohort.
- Developed RAG-based NLP applications using embedding models and vector databases for document retrieval and QA across 5+ use cases.
- Implemented text preprocessing, chunking, and similarity search pipelines, improving response relevance by 35% over baseline.

Cloud Services Trainee – National Telecommunication Institute (NTI)

Oct 2025 – Dec 2025

- Gained hands-on experience with 8 core AWS services including EC2, S3, IAM, VPC, Lambda, RDS, and CloudWatch.
- Designed secure, scalable cloud architectures aligned with AWS Solutions Architect best practices.
- Developed cost-optimized IAM policies and networking configurations, reducing projected cloud costs by 20%.

EDUCATION

B.Sc. in Computer Science – Major in AI and Machine Learning

2022 – 2026

Benha National University | Graduating GPA: 3.67 / 4.0

PROJECTS

Local LLM-Powered Research Assistant Chatbot

- Built a RAG-based research assistant chatbot using LangChain and Streamlit, supporting ingestion of 50+ PDFs simultaneously.
- Implemented PDF ingestion and document chunking using PyPDFLoader and RecursiveCharacterTextSplitter, reducing chunk retrieval time by 40%.
- Used Ollama LLaMA3 embeddings with Chroma vector database for semantic retrieval, achieving 88% answer relevance on test queries.
- Designed conversational interface with persistent chat history using Streamlit session state.

Cancer Information Chatbot

- Developed an interactive medical information chatbot using LangChain and Streamlit, indexing 200+ medical documents for QA.
- Built RAG pipelines with FAISS vector database and local embeddings, delivering accurate responses in under 2 seconds per query.
- Optimized performance using vector store caching, reducing repeated query latency by 60%.

Boycotted Products Recommendation System

- Built a recommendation system supporting 100+ products using Sentence Transformers embeddings and K-Means clustering.
- Implemented barcode scanning and product recognition using YOLO and OpenCV, achieving 94% detection accuracy.
- Improved recommendation relevance by 30% using embedding-based similarity matching.
- Deployed the system using Docker containers on Microsoft Azure, handling 500+ concurrent product lookups.

CERTIFICATIONS

AWS Certified AI Practitioner | AWS Certified Machine Learning Associate | AWS Certified Solutions Architect Associate

SKILLS

LLMs & Agentic AI: Hugging Face Transformers, LangChain, LangGraph, CrewAI, RAG, n8n, Prompt Engineering, Fine-tuning
Programming: Python (Advanced), C++ (Intermediate), C# (Beginner)
Orchestration & Deployment: FastAPI, Streamlit, Docker, AWS
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn
Databases: MySQL, Neo4j, MongoDB

LANGUAGES

Arabic – Native | English – Intermediate